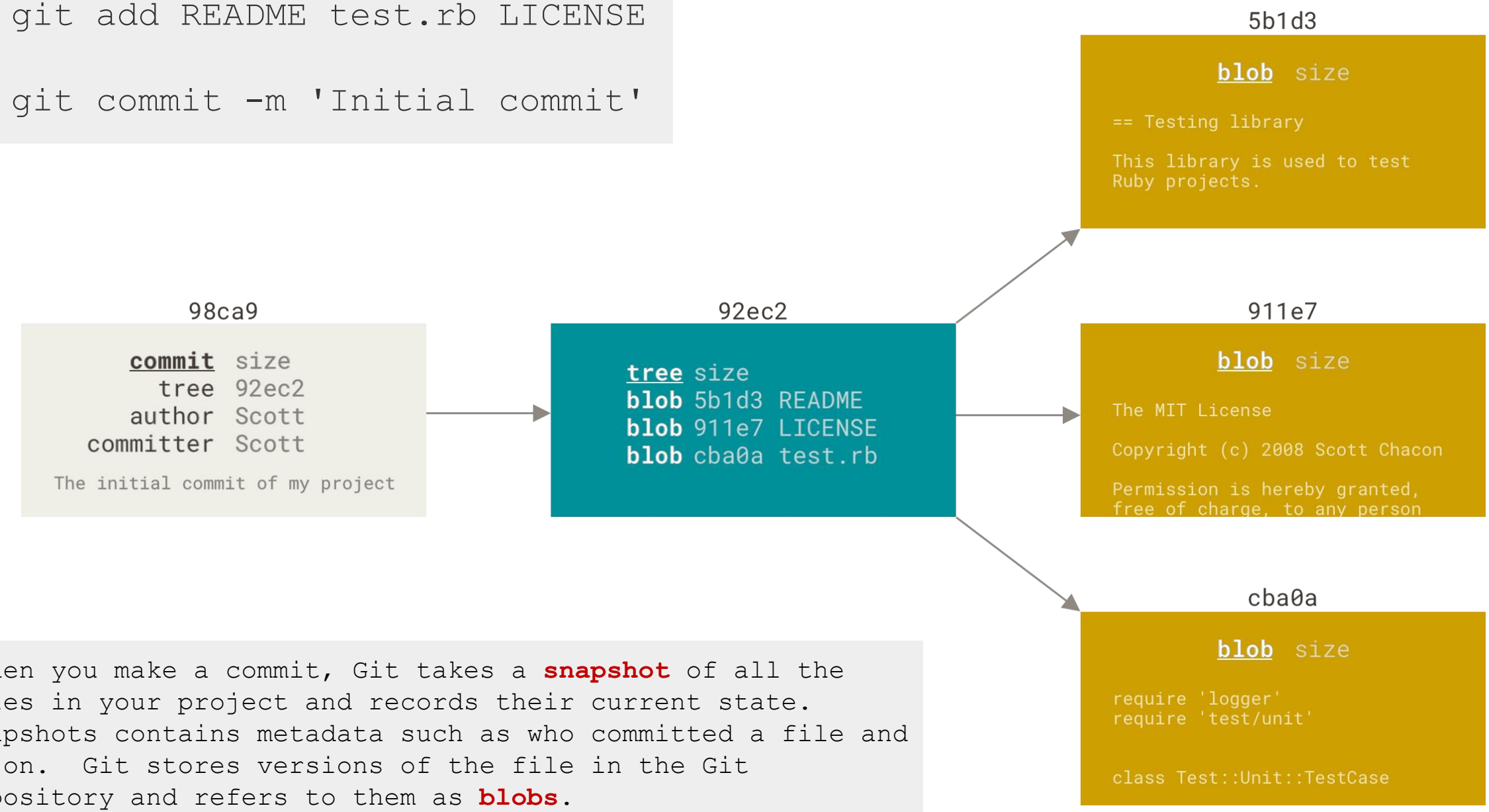
Branching and Merging



```
$ git add README test.rb LICENSE

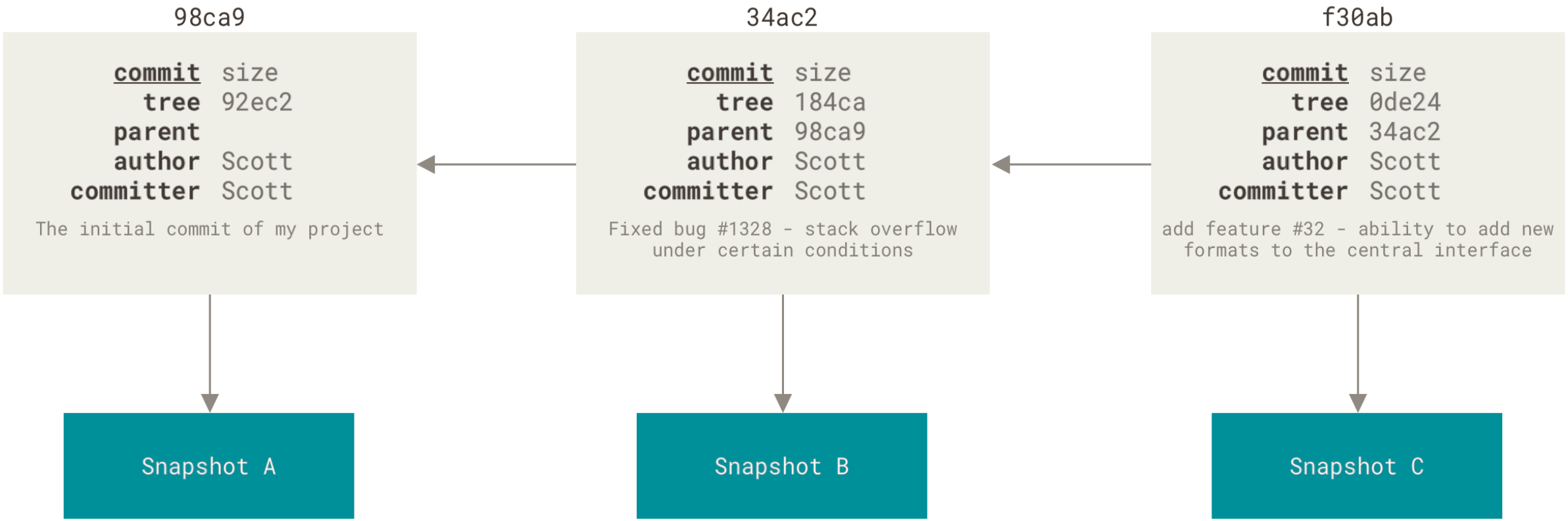
$ git commit -m 'Initial commit'
```



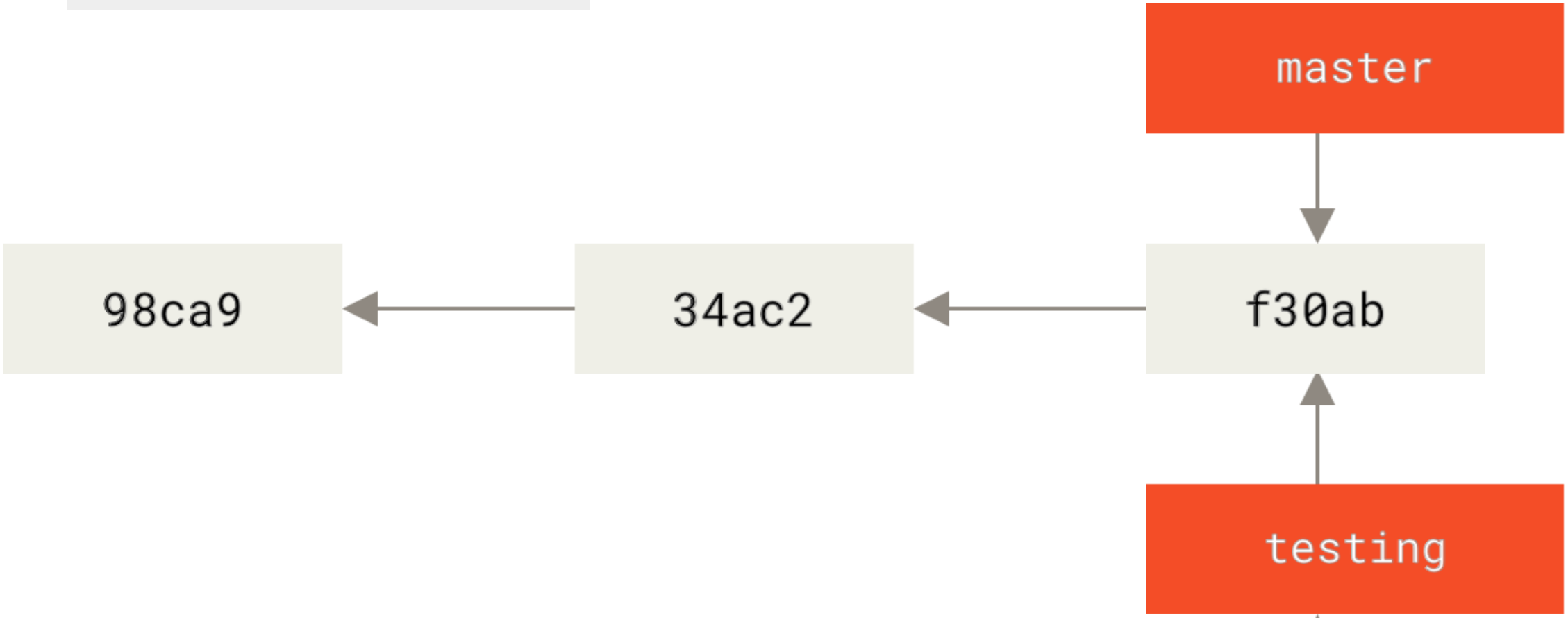
When you make a commit, Git takes a **snapshot** of all the files in your project and records their current state. Snapshots contains metadata such as who committed a file and so on. Git stores versions of the file in the Git repository and refers to them as **blobs**.

```
$ git commit -m 'B'

$ git commit -m 'C'
```

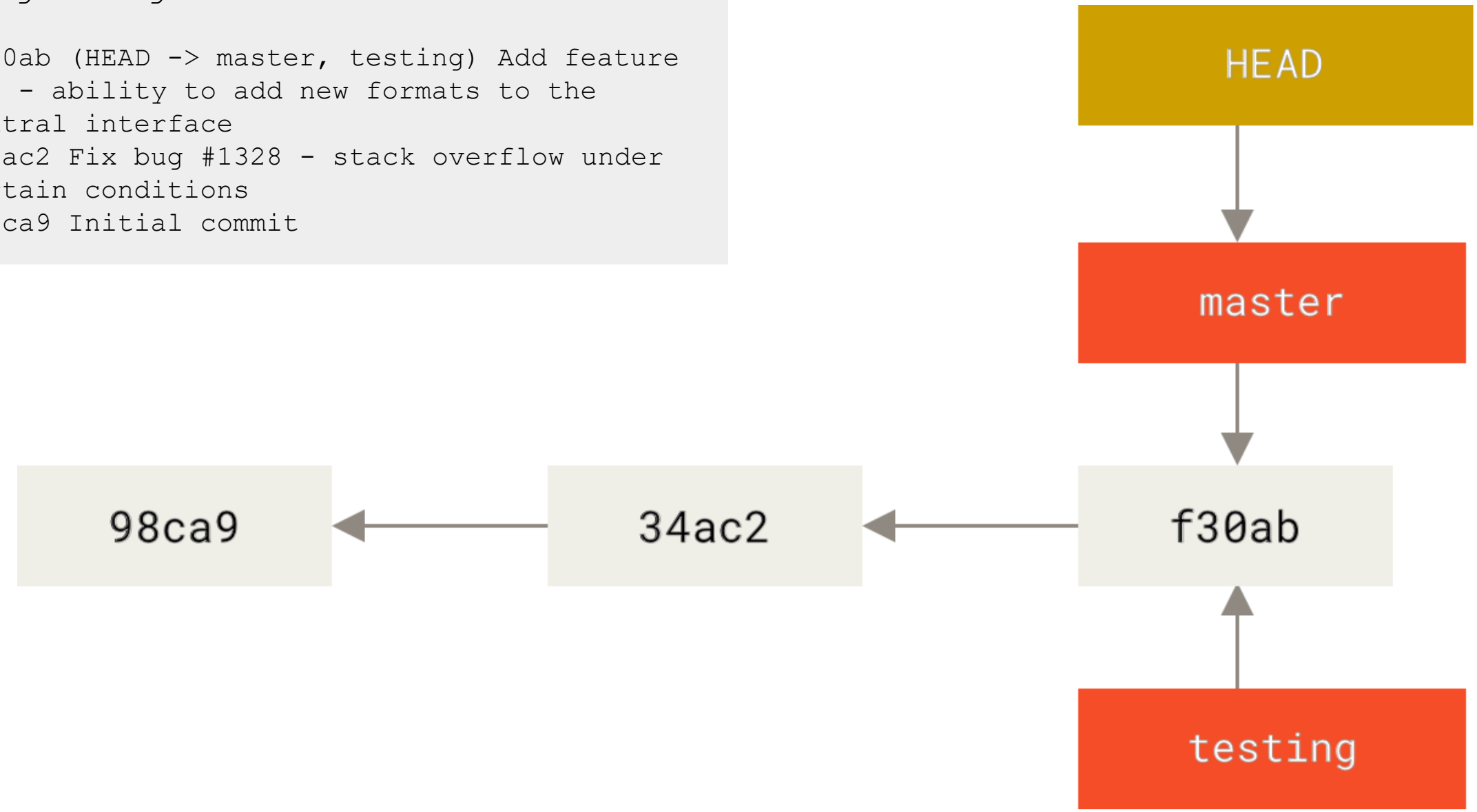


```
$ git branch testing
```

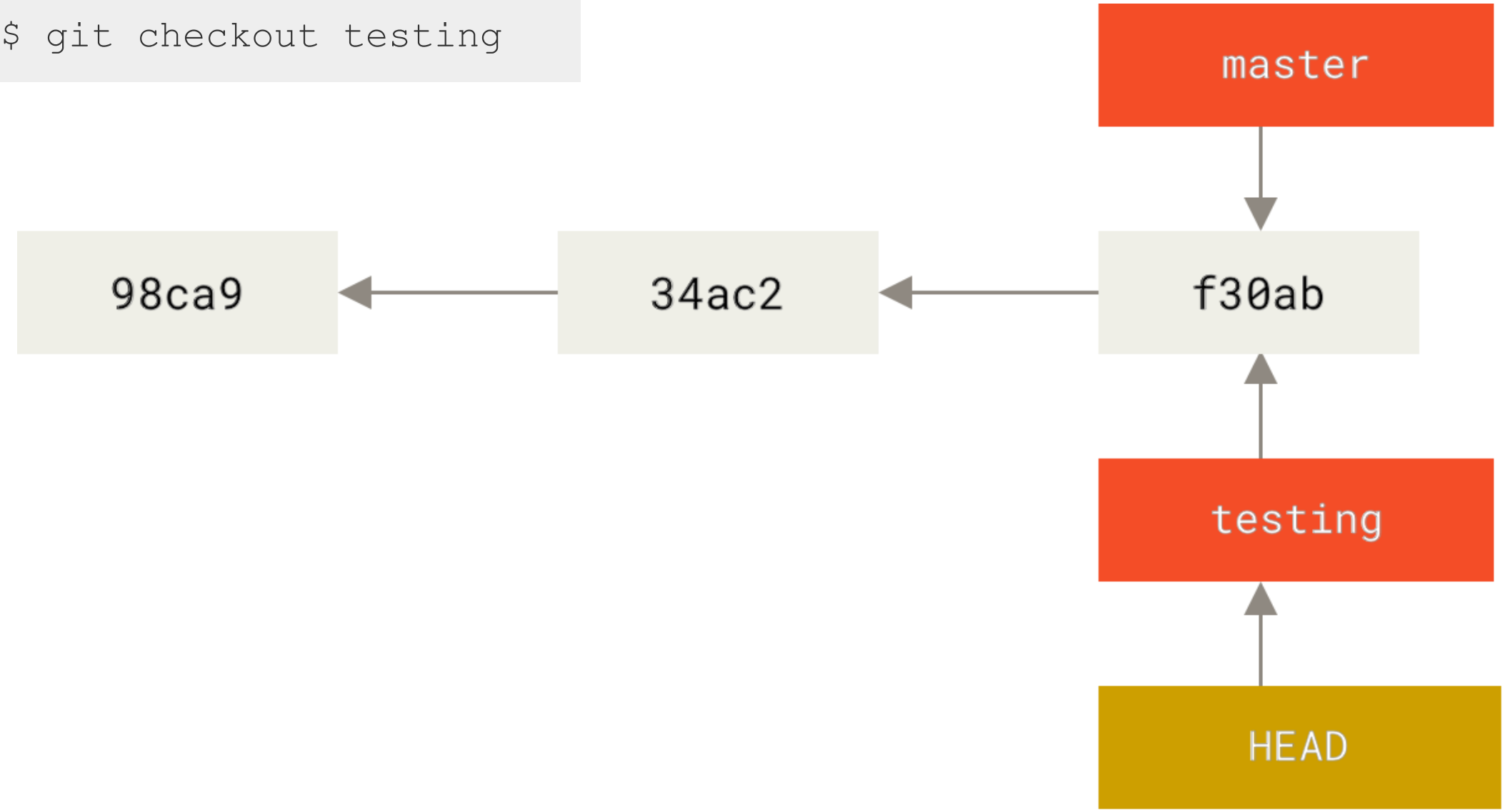


```
$ git log --oneline -decorate

 f30ab (HEAD -> master, testing) Add feature
#32 - ability to add new formats to the
central interface
 34ac2 Fix bug #1328 - stack overflow under
certain conditions
 98ca9 Initial commit
```

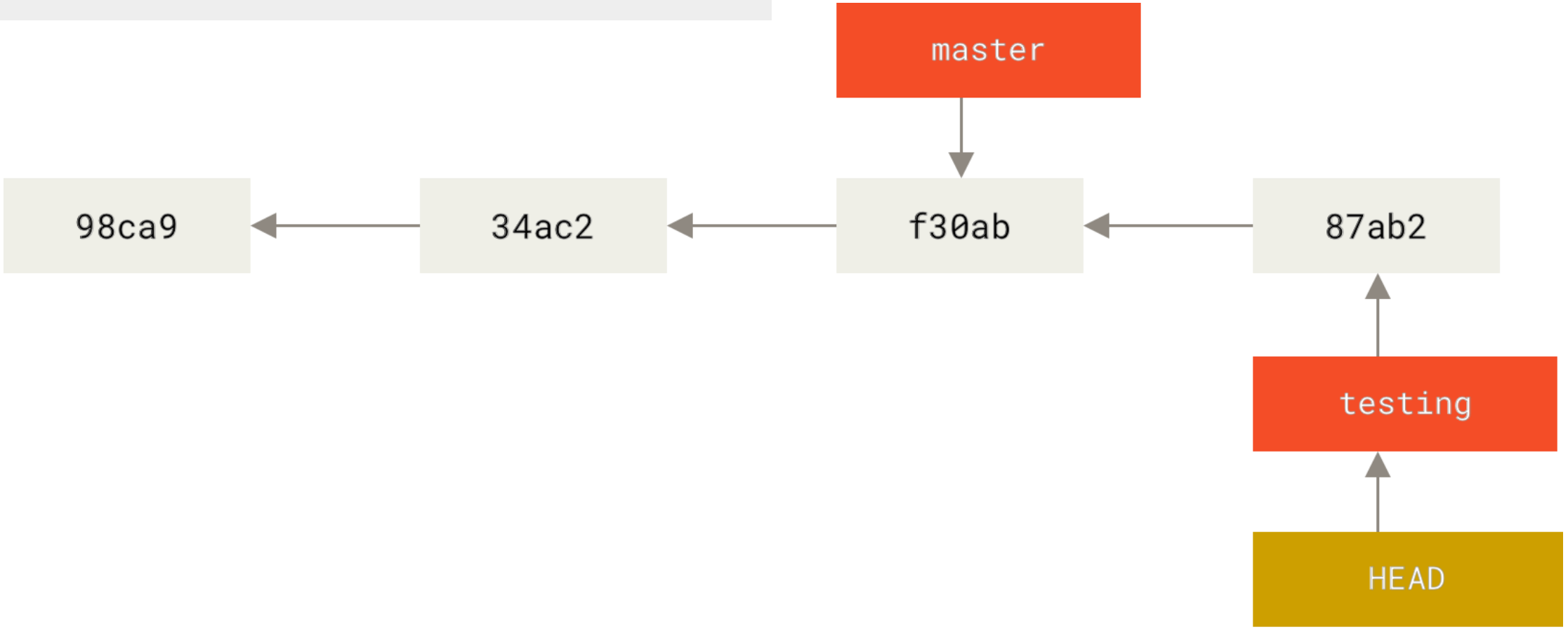


```
$ git checkout testing
```

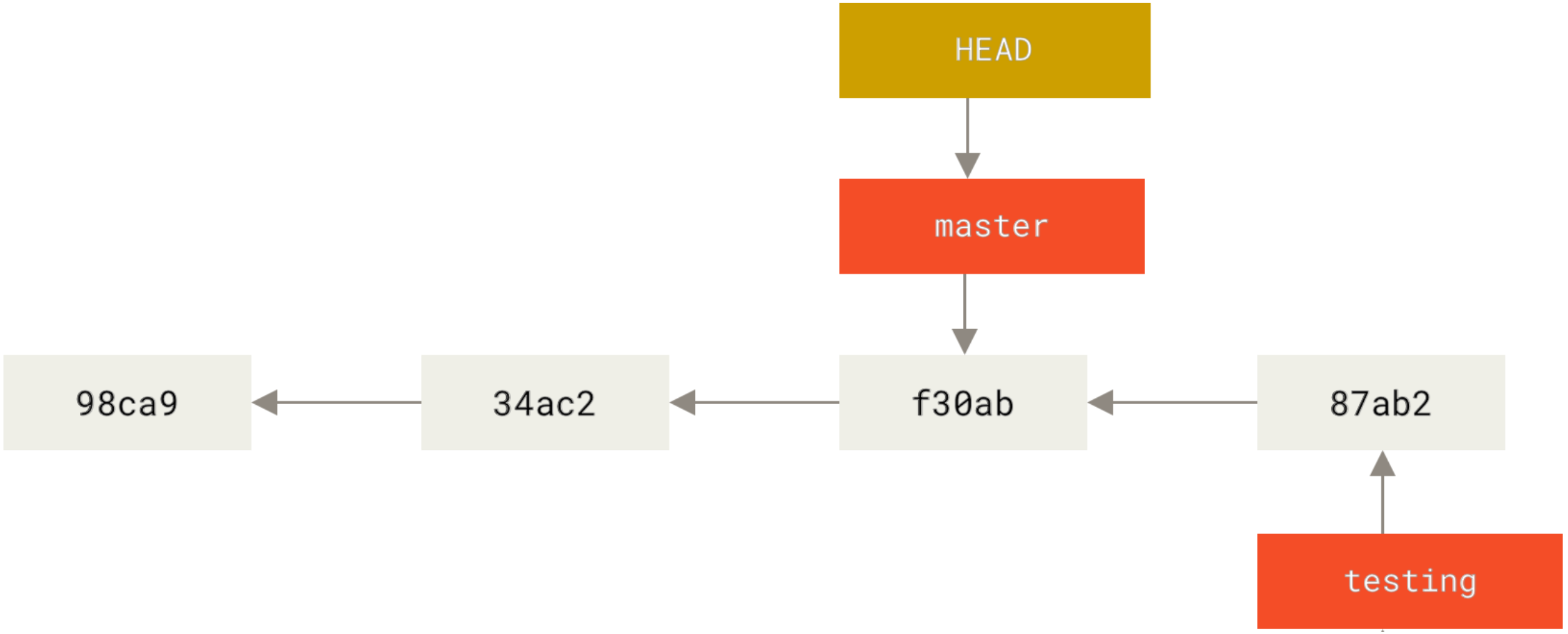


```
$ sudo nano test.rb

$ git commit -a -m 'Make a change'
```

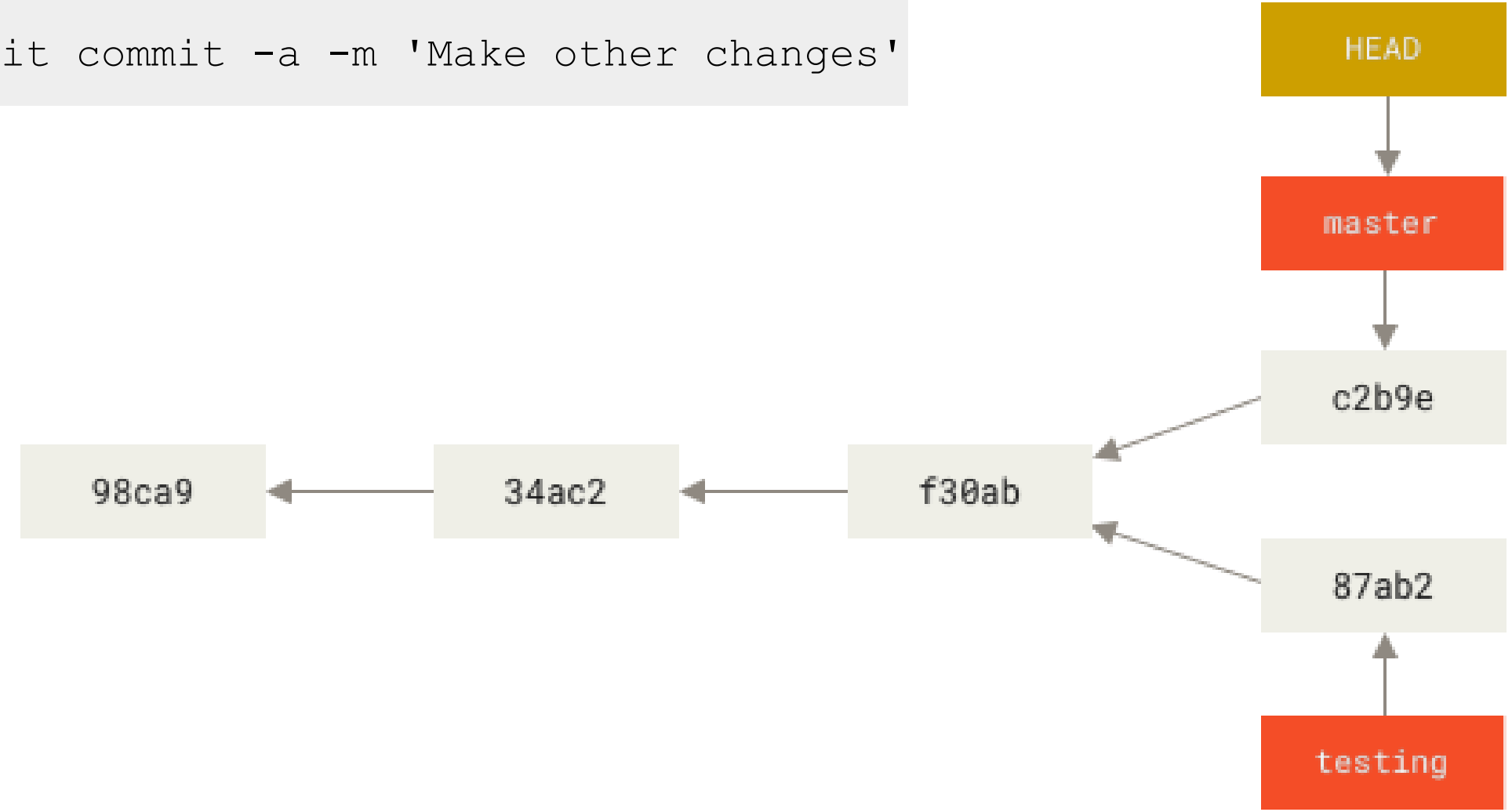


```
$ git checkout master
```



```
$ sudo nano test.rb

$ git commit -a -m 'Make other changes'
```

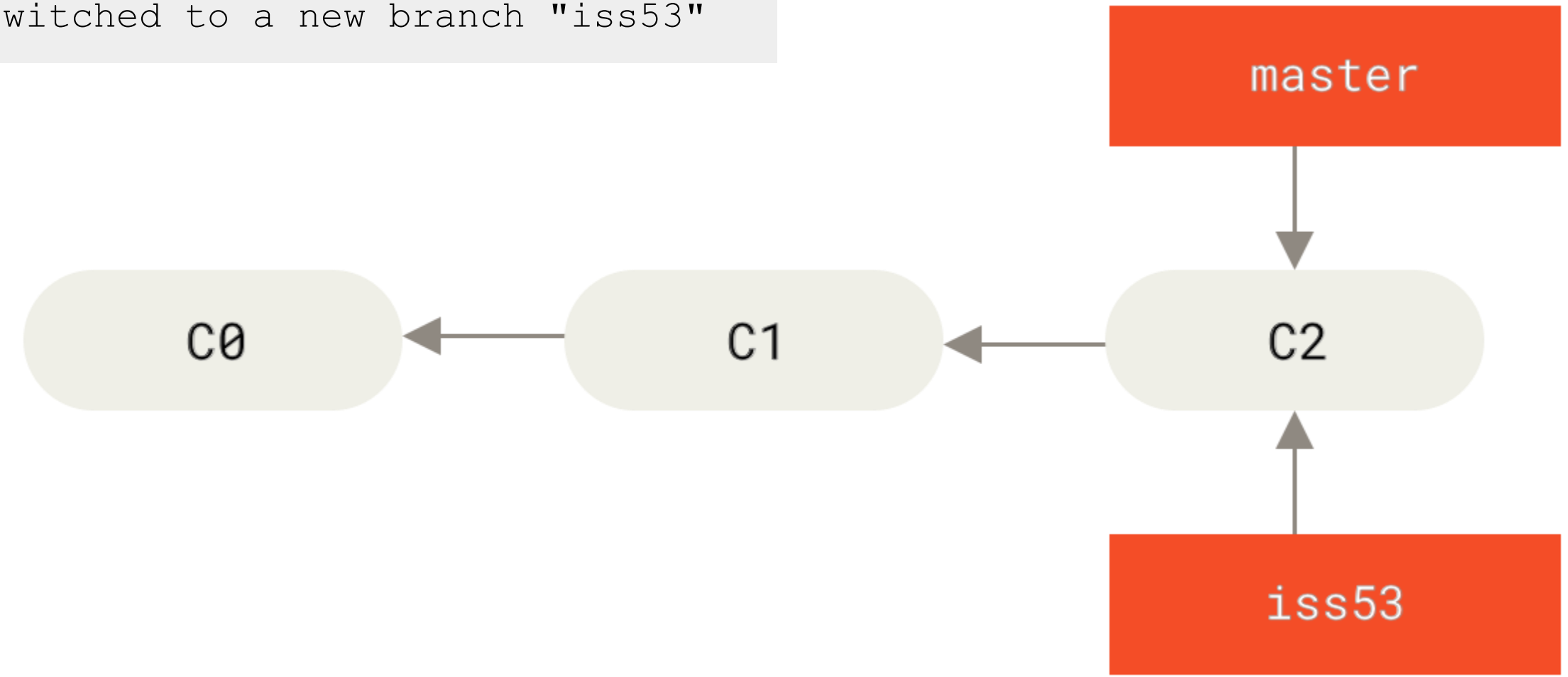


Merging with a scenario

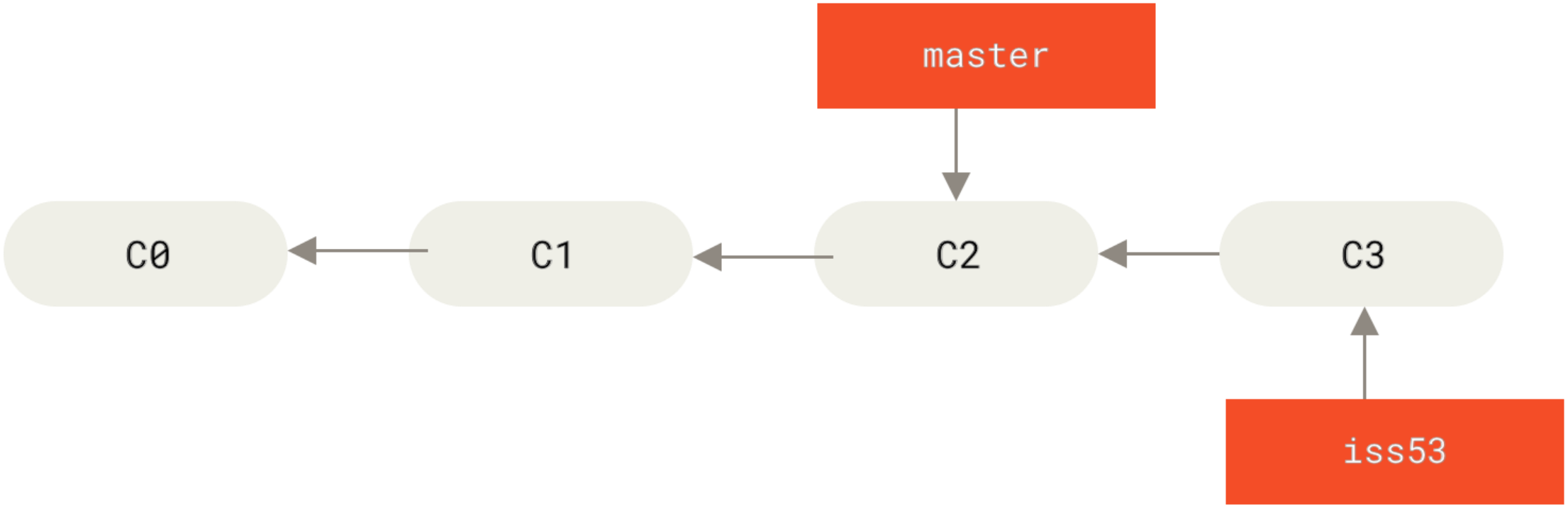



master
C0
C1
C2

```
 $ git checkout -b iss53
Switched to a new branch "iss53"
```



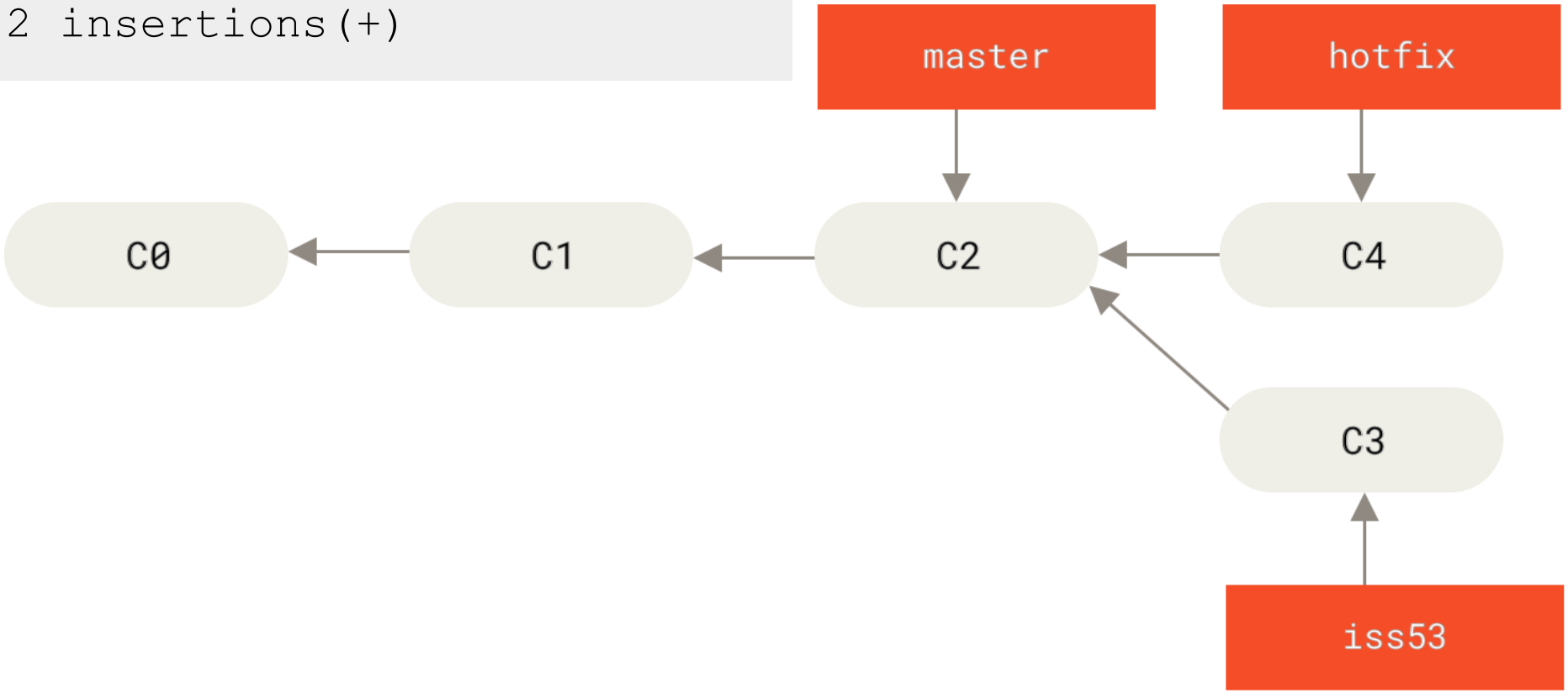
```
$ sudo nano index.html
$ git commit -a -m 'Create new footer [issue 53]'
```



```
 $ git checkout -b hotfix
Switched to a new branch 'hotfix'

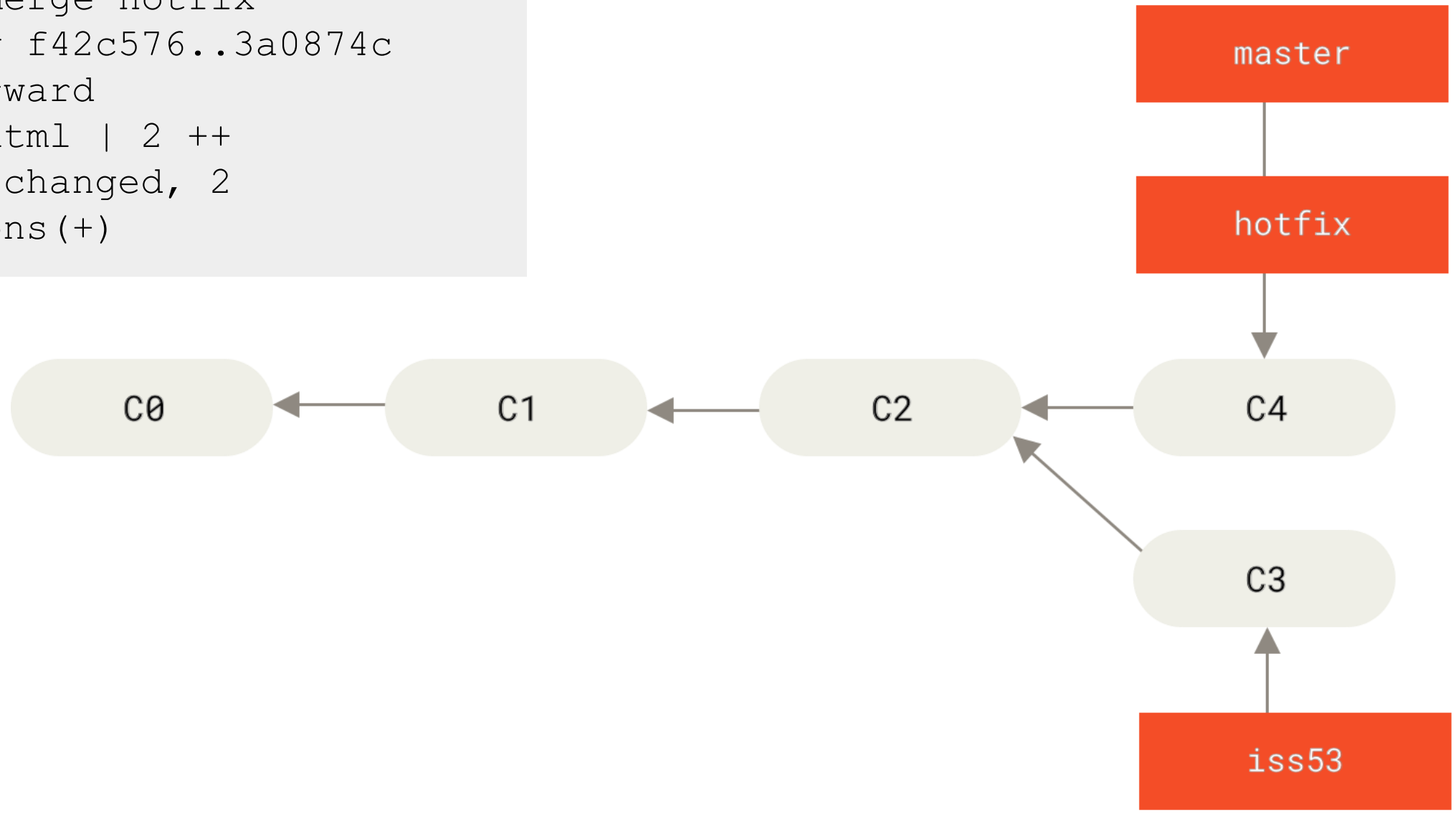
 $ sudo nano index.html

 $ git commit -a -m 'Fix broken email address'
[hotfix 1fb7853] Fix broken email address
 1 file changed, 2 insertions(+)
```



```
 $ git checkout master

 $ git merge hotfix
Updating f42c576..3a0874c
Fast-forward
 index.html | 2 ++
 1 file changed, 2
insertions(+)
```



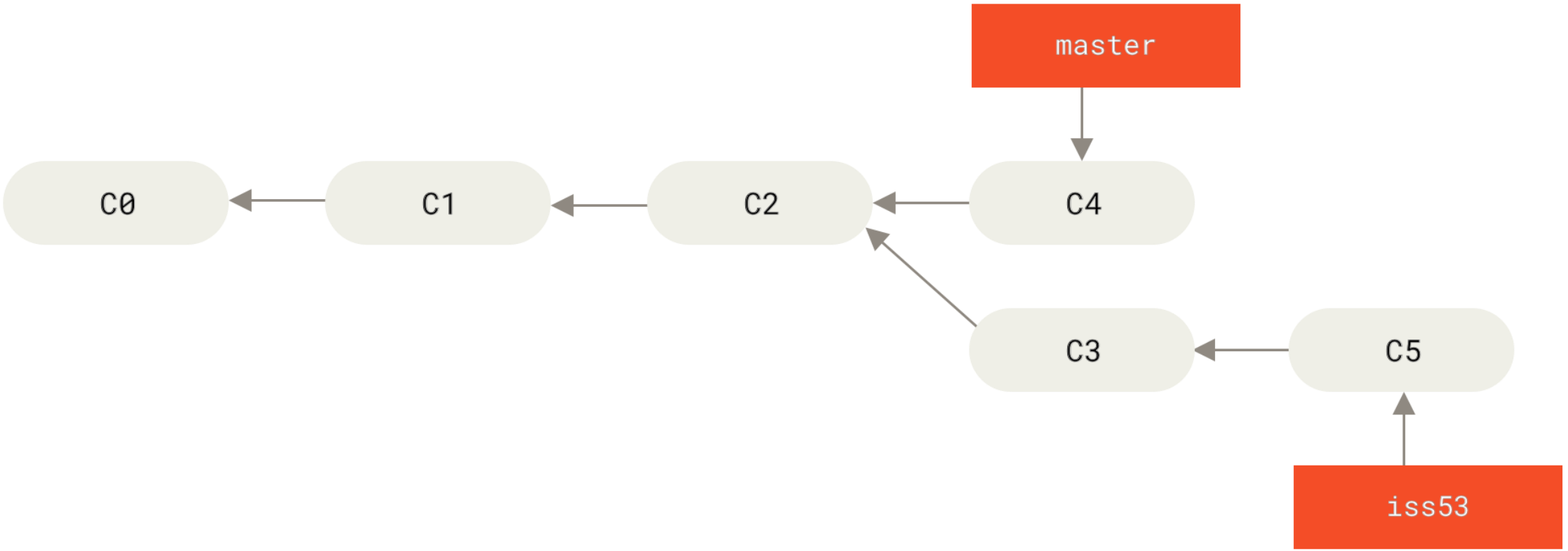
```
$ git branch -d hotfix

Deleted branch hotfix (3a0874c).
```

```
 $ git checkout iss53
Switched to branch "iss53"

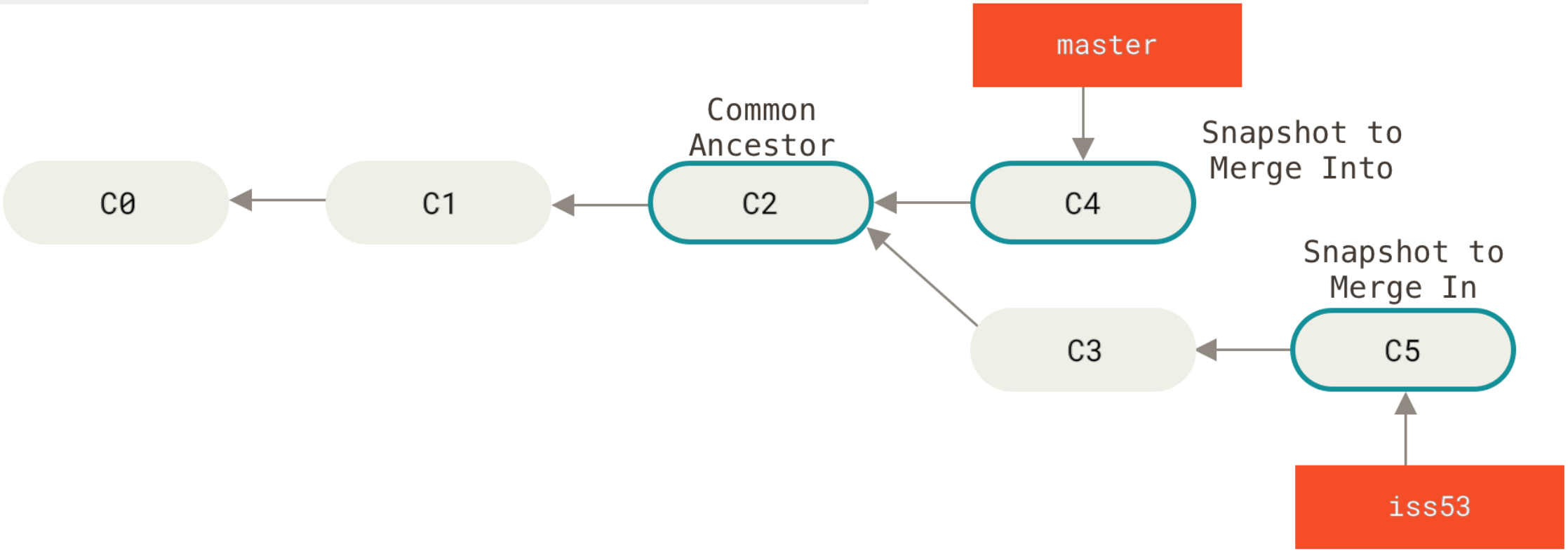
$ sudo nano index.html

$ git commit -a -m 'Finish the new footer [issue 53]'
[iss53 ad82d7a] Finish the new footer [issue 53]
1 file changed, 1 insertion(+)
```

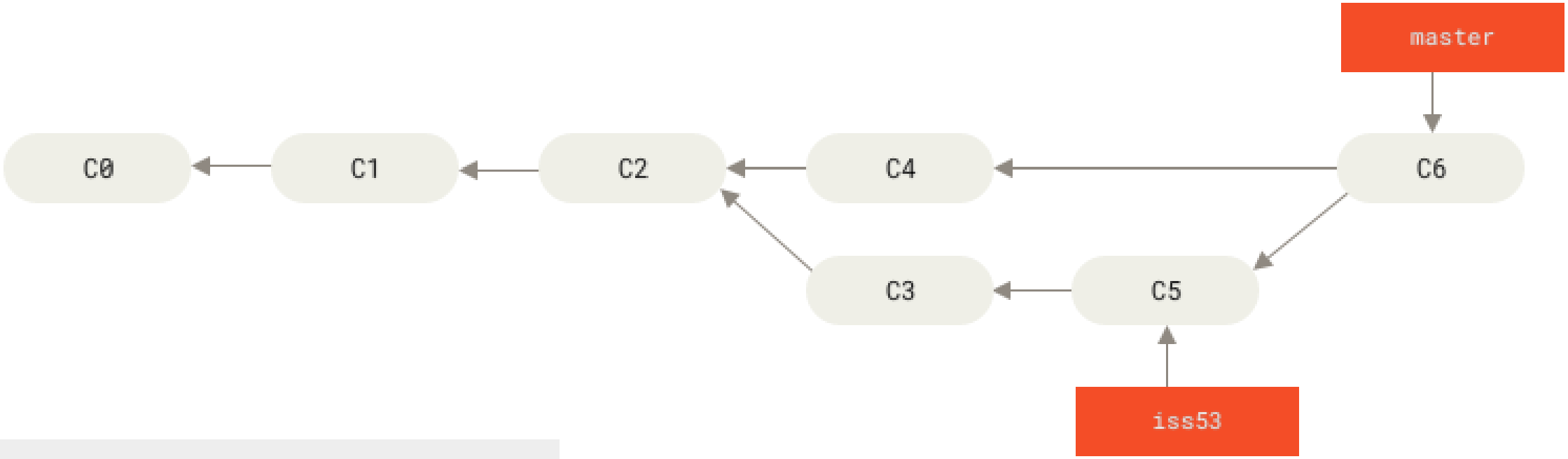


```
 $ git checkout master
Switched to branch 'master'

 $ git merge iss53
Merge made by the 'recursive' strategy.
index.html |    1 +
1 file changed, 1 insertion(+)
```



Instead of just moving the branch pointer forward, Git creates a new snapshot that results from this three-way merge and automatically creates a new commit that points to it. This is referred to as a merge commit, and is special in that it has more than one parent.




```
$ git branch -d iss53
```



1. See this Repo
2. Reference 1
3. Reference 2
4. Rebasing- For your future